Backend Developer Skills: Definition And Examples

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Backend developers are responsible for designing, developing and maintaining the inner components, aka the server-side of web applications. They primarily work on the databases, servers and application program interfaces (APIs) that run behind the scenes to make sure that the client-side of a website remains operational. Due to the technical nature of their job role, backend developers need some highly specialised skills.

In this article, we discuss and provide examples of backend developer skills, explain how to improve your skills and suggest how to highlight them during the hiring process.

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What Are Backend Developer Skills?

Backend developer skills are a combination of technical and soft skills required to perform all backend developer duties successfully. As a backend developer, you should have the technical skills to work with server-side scripting languages, APIs and database management tools. You also need soft skills like communication, problem-solving and creativity to build a successful career in backend development.

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Examples Of Backend Developer Skills

Here are some examples of backend developer skills:

Technical skills

To become a successful backend developer, focus on these four technical skills:

Programming and server languages

Backend developers write code that helps build and maintain the technology that powers the frontend side of a web application. That is why they should be proficient in programming languages like Java, Python, .NET, PHP and Ruby. You do not need proficiency in all of these programming languages, but you should know at least one language. Based on your interest area, choose a language you want to learn and master.

Database management

One of the essential job responsibilities of a backend developer is to manage a hosting environment. Database management requires in-depth knowledge of database administration and database management systems (DBMS). Developers should be familiar with recovering and stacking data from databases. Also, they create storage solutions in the application database to store a large amount of information. Some commonly used database programs include MySQL and MongoDB.

Knowledge of API

APIs are a set of functions that help create an application by connecting it to the internet and sending data to the server. The server retrieves and interprets the data and performs the desired action on it. It then sends the data back to the application. Finally, the application presents the data to you in a readable format. The entire process happens via an API. For this reason, backend developers should know how to write API codes that create error-free mobile or web applications.

Knowledge of middleware

Middleware is any software on a server that helps to connect the application's frontend with the backend. Backend developers need to know middleware as it controls the flow of information between the database, operating system, application and server. Good middleware increases IT efficiency and enhances business process management.

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Soft skills

Here are four important soft skills an employer looks for in a backend developer:

Communication

The ability to interact with your colleagues and stakeholders to ensure the final product meets the project's specification requires excellent verbal communication skills. Also, a large part of your work may involve documenting the code or design and writing test cases. This requires mastery of written communication skills. When you document your work, it becomes easier for other developers and quality analysts to make changes as per project requirement.

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Active listening

Backend developers often need to attend meetings with clients and stakeholders to understand the project's requirement. Having active listening skills will help them build connections with clients, win their trust and avoid missing any crucial information.

Problem-solving

Problem-solving skills help you become an efficient developer. Backend developers work on a range of tasks like coding, designing APIs and troubleshooting. They spend a large portion of their workday detecting bugs in the code causing the program to break. Backend developers should find solutions to different programming obstacles like debugging code, making revisions to a website without altering the functionality and preventing unexpected website crashes.

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Technical creativity

Technical creativity can be useful for backend developers as it helps them design the look of a website and structure the code. Also, they may have to use their creativity to find out ways to structure a database to model complex project requirements. Another aspect of creativity is the ability to write codes that other developers and quality analysts can easily understand.

How To Improve Backend Developer Skills?

Here are four ways to improve your backend developer skills:

1. Practise coding

As backend developers write codes to build communication between database and application, you need to practise coding in one of the programming languages like Java, PHP, Python or Ruby based on the job you are applying to. The more you practise coding, the more experienced backend developer you can become. You can practise it using free online software.

2. Learn new programming languages

If there is a programming language you do not have experience in, enrol in a programming language learning course. For instance, if you know PHP but have no experience in Python, enrol in an online Python course. It helps you come out of the comfort zone and you learn a new skill that may be useful in your job.

3. Learn frontend languages

Backend developers ensure that they deliver data and systems requested by the frontend application as per specifications. But, when the application does not meet the specification, you may have to communicate and collaborate with a frontend developer. So, it becomes essential to know the frontend languages like CSS, HTML and JavaScript. When you expand your skill set, it becomes easier to communicate and find a solution to a problem. You can always learn these frontend languages using open-source websites.

4. Read books

To advance your career and become a better backend developer, read books. You can either read technical books to improve your technical knowledge or read self-help books to improve your soft skills. When you read books, you understand things from a unique perspective, learn from the mistakes of others and can implement the knowledge gained in your workplace.

How To Highlight Backend Developer Skills?

To get a backend development job, you should highlight your skills on your CV, profile summary and during an interview. Here are three ways to highlight your backend developer skills:

1. In your CV

In your CV, under the skills section, mention the backend developer skills and your level of proficiency in them. It makes your CV stand out and increase your chances of qualifying for the next round. You can even mention these skills in the experience section of your CV. Regardless of where you mention these skills, include keywords related to the job description. Using keywords is essential because employers typically use an applicant tracking system (ATS) to shortlist candidates and the software does keyword scoring to find best-fit candidates.

You can add the backend developer skills in your skills section in the following way:

- *4+ years of experience using version control and source code management tools*
- *Strong communication and problem-solving skills*
- *Expert knowledge of database technologies*
- *Excellent coding skills*

You can add the backend developer skills in your experience section in the following way:

- *Migrated API written in ES version 1.9 to ES version 4.x*
- *Developed new and maintained existing applications*
- *Implemented algorithms and solved system-related problems*
- *Delivered projects using agile methodology*
- *Delivered seminars to the team around new backend development approach technologies and workflow*
- *Wrote re-useable test documents to ensure 100% code coverage*
- *Worked in time-sensitive and complex projects*

- *Discussed project requirements with stakeholders to meet project specification*

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2. In your profile summary

Your profile summary is an excellent place to showcase your backend developer skills. Try to use keywords like “Java”, “API” or “SQL” and action verbs to express your proficiency in a particular backend skill. This helps make your profile summary stand out.

When writing your profile summary, use the following example as an inspiration:

“Passionate and creative backend developer with five years of experience designing, building and enhancing .Net web application for EdTech companies. Proficient in SQL, APIs and GraphQL.”

3. In your interview

Interviews help you display your qualities and experience to the employer and gives you a chance to elaborate on your CV. Some employers ask behavioural questions to understand your personality, while some ask technical questions to learn about your technical skills and experience in a similar job role. Questions on technical challenges help you prove your problem-solving and backend developer skills.

Navigating Modern Backend Technologies

Backend developers today work with an expanding ecosystem of technologies beyond basic programming languages. Expertise in frameworks like Django and Node.js has become increasingly valuable, as has proficiency with DevOps tools including Git, GitHub, and containerization solutions such as Docker and Kubernetes. Understanding cloud computing platforms from Amazon AWS, Microsoft Azure, and serverless architectures enables developers to build scalable applications with optimal performance. Knowledge of both relational databases and NoSQL databases like PostgreSQL and MongoDB is expected, while experience implementing RESTful APIs and authentication systems remains fundamental to backend development. A comprehensive roadmap for professional growth should include regular exposure to these evolving technologies.

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The Path To Backend Excellence

For aspiring backend developers, building a strong foundation in computer science and data structures provides the groundwork for handling complex development processes. Many professionals enhance their careers through relevant certifications and by learning front-end technologies to better collaborate with team members. The development landscape increasingly demands understanding of microservices architecture and version control systems as projects grow in complexity. Software engineering practices that optimize user experience through efficient APIs have significant impact on overall application quality. As organisations embrace cloud computing and containerization, backend developers who can navigate both software development fundamentals and emerging web frameworks find themselves well-positioned for career advancement with competitive developer salary packages across the technology sector.

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